

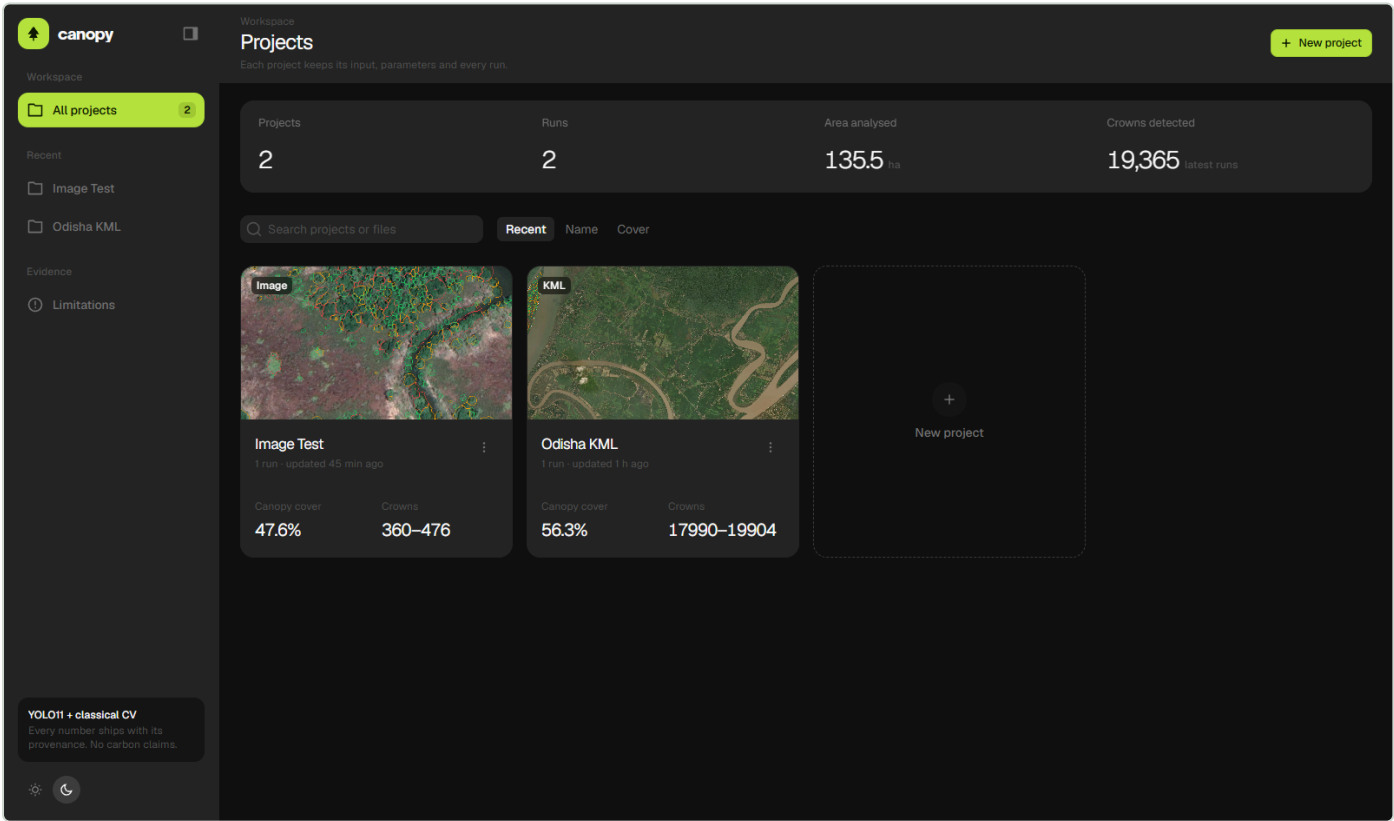
CANOPY: tab-by-tab guide

What each screen shows, what you can do on it, and one line to say about it in the demo.

Screen	In one sentence
Projects	Every workspace, with a thumbnail of its latest crowns, its cover and its crown range.
New workspace	Upload a file (KML, KMZ, GeoJSON, GeoTIFF, PNG, JPG), draw an area, or use the sample, then review before anything is created.
Delete workspace	Type the workspace's name to confirm, so nothing is deleted by accident.
Overview	The headline numbers: cover, crowns, plausible range, confidence, height, warnings and downloads.
Map + Parameters	Crowns on the imagery, plus every setting (detector, AI model size, crown size, threshold) with one-click re-runs.
Crowns	A table of every crown with its measurements and why it got its confidence score.
Pipeline	The 7 processing stages as images you can step through, each with its method and numbers.
Validation	Click real trees in a small patch to measure precision and recall on your own site.
Runs	History of every run, what changed between them, and the results side by side.
Audit	Downloads (GeoJSON, CSV, overlay, audit zip) and the manifest of how the numbers were made.
Limitations	What the tool cannot measure, stated plainly.
Ask AI	A per-workspace assistant that answers from this workspace's data and can propose a better run.

1 Projects

sidebar → All projects



What it shows

- Totals across all workspaces: projects, runs, area analysed, crowns detected.
- One card per workspace: input type badge (KML, Image, GeoTIFF...), a thumbnail with crowns drawn on it, canopy cover and crown range.

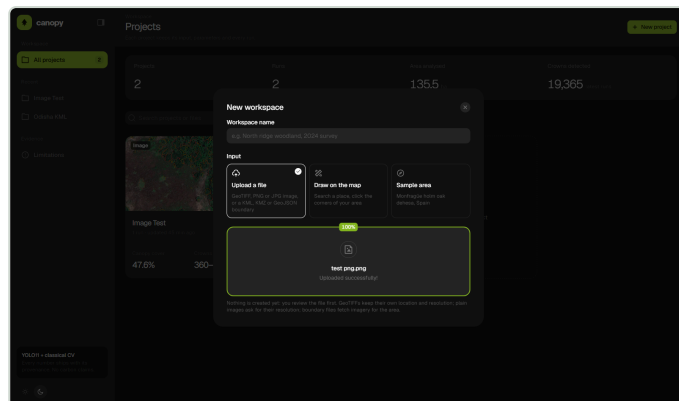
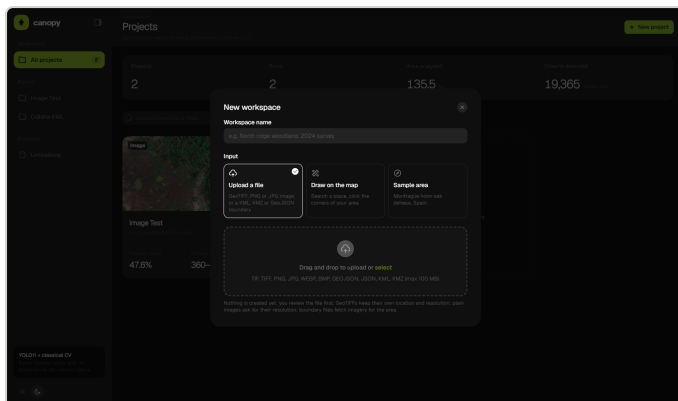
What you can do

- Search by name or file, sort by recent, name or cover.
- Open, rename or delete a workspace from the : menu; start a new one.

Say: "Each workspace keeps its input, its parameters and every run, so the work is never lost and always reproducible."

2 New workspace

New project button



What it shows

- Three ways in: **upload a file**, **draw on the map**, or the **sample area**.
- After upload, a **review step**: file type, size, pixel dimensions, number of polygons, and what will happen next.
- For a plain PNG/JPG (no location): a **ground resolution** field with presets: Drone 0.03 m, Aerial 0.1 m, Satellite HD 0.3 m, Satellite 0.5 m, Coarse 1 m, Sentinel-2 10 m.

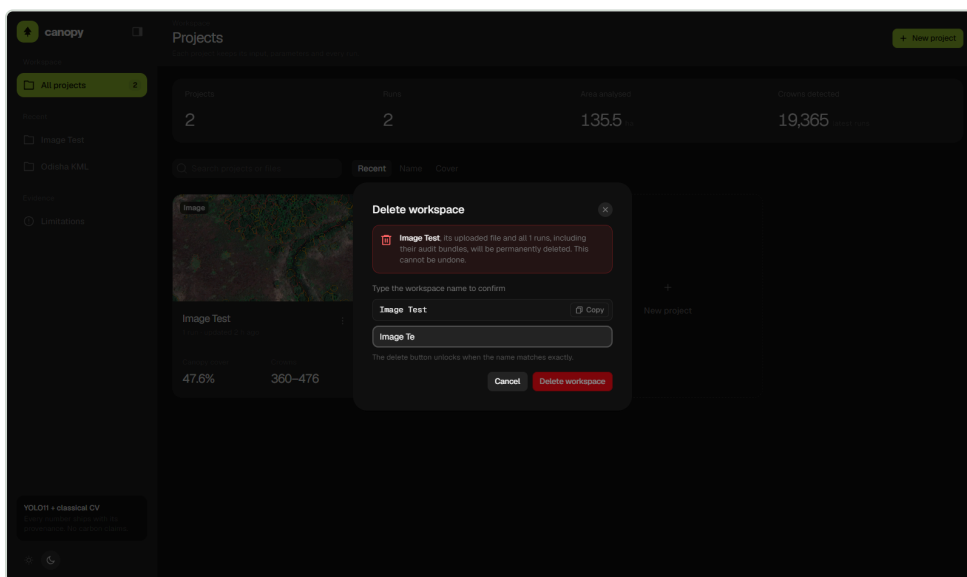
Say: "A screenshot has no scale, so CANOPY asks for it instead of guessing; every area and diameter is then in real metres."

What you can do

- Pick which polygons of a multi-polygon KML to analyse (select all, or tick a few).
- Change the file, or cancel; nothing is created until you press **Create workspace**.
- A growing-folder animation plays while the workspace is created and the first run starts.

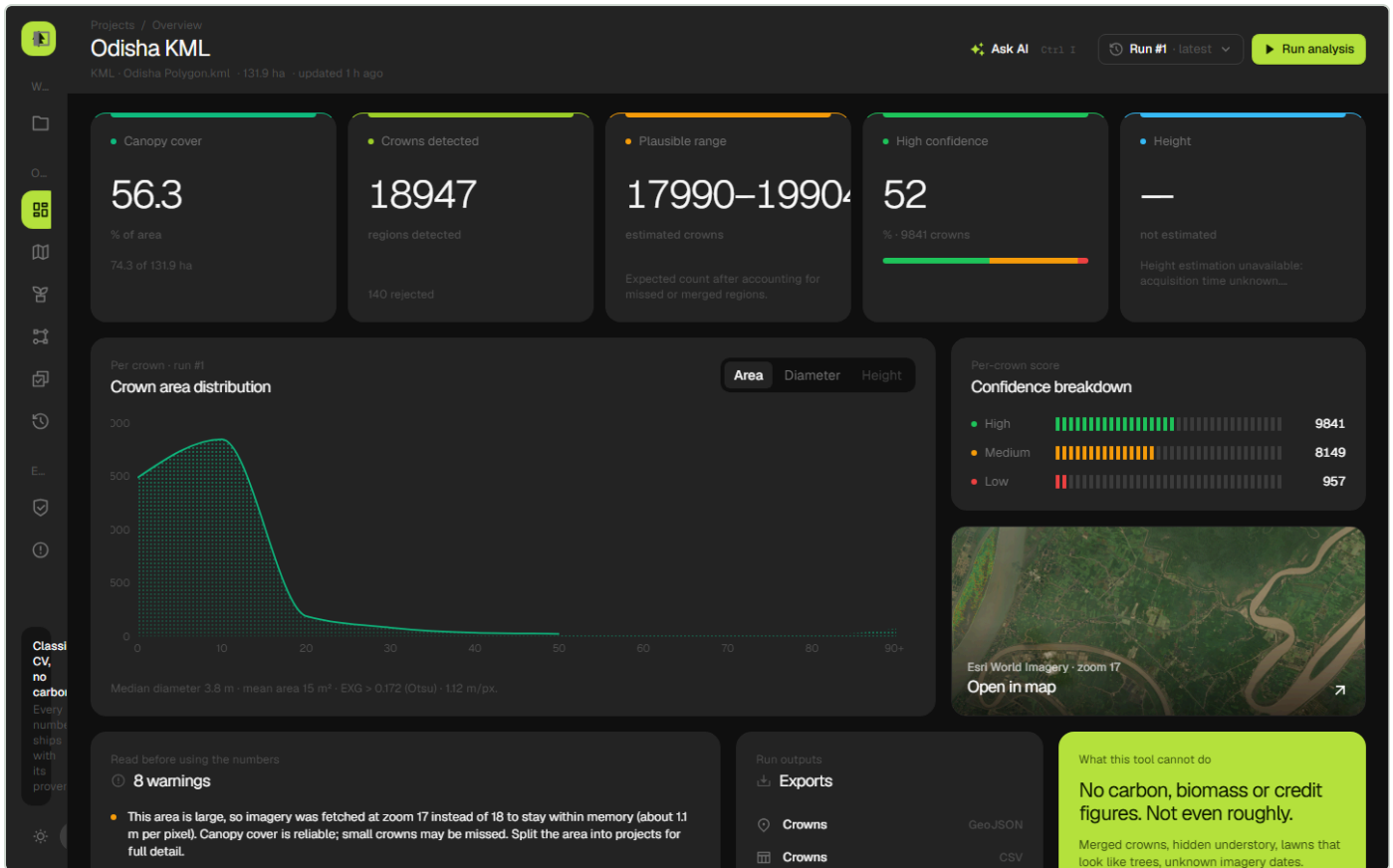
3 Delete workspace

: menu → Delete



The dialog says exactly what will be lost (the uploaded file, every run and its audit bundle). The red button stays disabled until the workspace name is typed exactly.

Say: "Destructive actions need the name typed, the same pattern GitHub uses for deleting a repository."



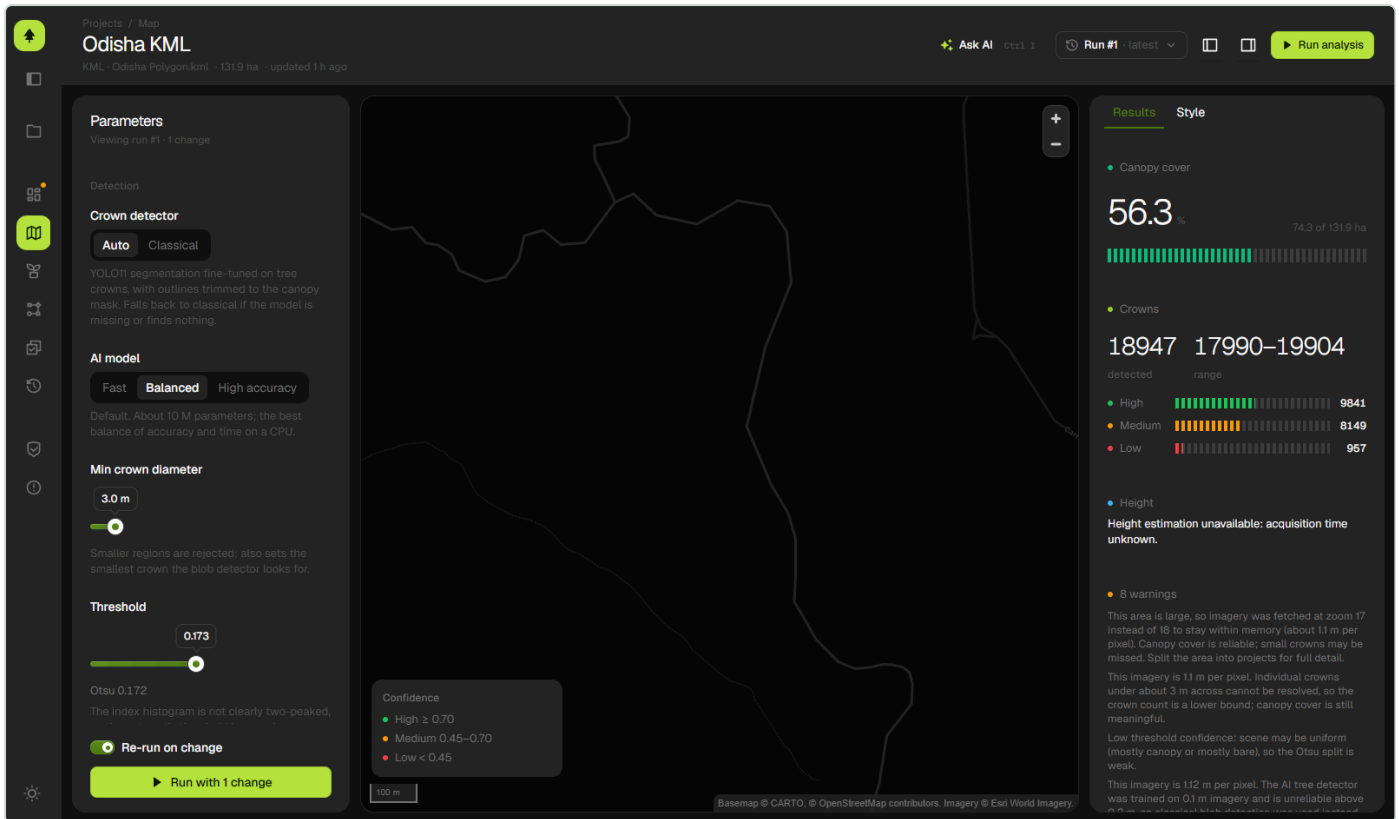
What it shows

- Canopy cover** (% of the area and hectares), **crowns detected**, the **plausible range**, the share of **high-confidence** crowns, and **height** (only when the capture time is known).
- Crown size distribution (area, diameter or height), the confidence breakdown, and a map preview.
- Warnings**: every assumption the run made (zoom lowered, resolution assumed, low-contrast threshold...).

What you can do

- Switch the chart between area, diameter and height.
- Download crowns (GeoJSON, CSV) from Exports.
- Small arrows show the change against the previous run.

Say: "The count is a range, not one number, because a low-confidence blob is often two merged trees. The warnings list every assumption behind it."



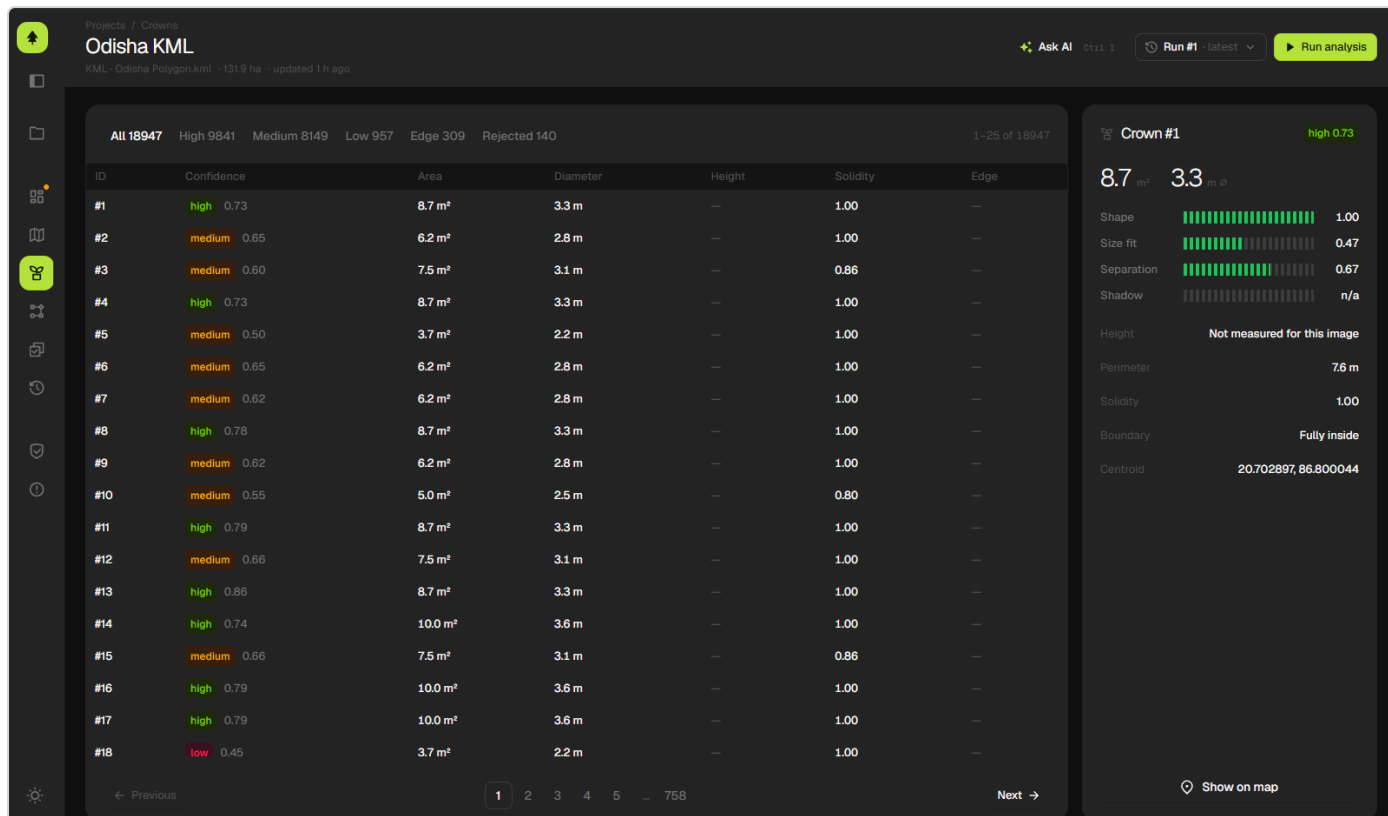
Left: Parameters

- **Crown detector:** Auto (AI) or Classical.
- **AI model:** Fast (YOLO11n), Balanced (YOLO11s, default), High accuracy (YOLO11m). Sizes that are not trained are greyed out.
- **Min crown diameter, vegetation index** (ExG, VARI, NDVI), **threshold** (Otsu or manual), **tile zoom**, capture time for height, and resolution for plain images.
- **Re-run on change:** the analysis re-runs by itself after you stop moving a slider.

Centre and right

- Map layers: imagery, canopy mask, crowns (coloured by confidence, area or height), and **rejected** regions, so nothing disappears silently.
- Click a crown for its measurements. The Style tab changes colours and fill.
- Right panel: results for the run being viewed.

Say: "You choose speed or accuracy: Fast for a quick look at a big area, High accuracy when the count matters. The run history keeps both."



- Every crown: ID, confidence (high / medium / low and the score), area, diameter, height, solidity, and whether it touches the edge.
- Filter chips: All, High, Medium, Low, Edge, Rejected. Click a row to see the **four signals** behind its score (shape, size against its neighbours, separation, shadow, plus the AI detector's own score).

Say: "Every crown can explain its own confidence score. Nothing is a black box."

canopy Projects / Pipeline Odisha KML
KML: Odisha Polygon.kml · 131.9 ha · updated 1 h ago

Workspace

All projects

Odisha KML

Overview 8

Map

Crowns 18947

Pipeline

Validation

Runs 1

Evidence

Audit

Limitations

YOLO11 + classical CV
Every number ships with its provenance. No carbon claims.

Detector: blob markers + watershed

Pipeline

- 1 Source imagery
1.12 m/px
- 2 EXG vegetation index
pi-p99 0.0213 to 0.3651
- 3 Threshold
otsu at 0.172
- 4 Cleaned canopy mask
56.3% canopy cover
- 5 Crown centres
19087 crown centres**
- 6 Crown segments
19087 regions
- 7 Filtered and scored crowns
18947 crowns, range 17990–19904

Stage 5 of 7
Crown centres



Stage output over imagery

What happens here

Method

Sunlit crowns are brightest and greenest in the middle. A Laplacian-of-Gaussian blob detector, run at every crown size from half the minimum diameter up to the maximum, finds one peak per crown even where crowns touch. White dots are the crown centres that fall on canopy; each seeds one watershed region.

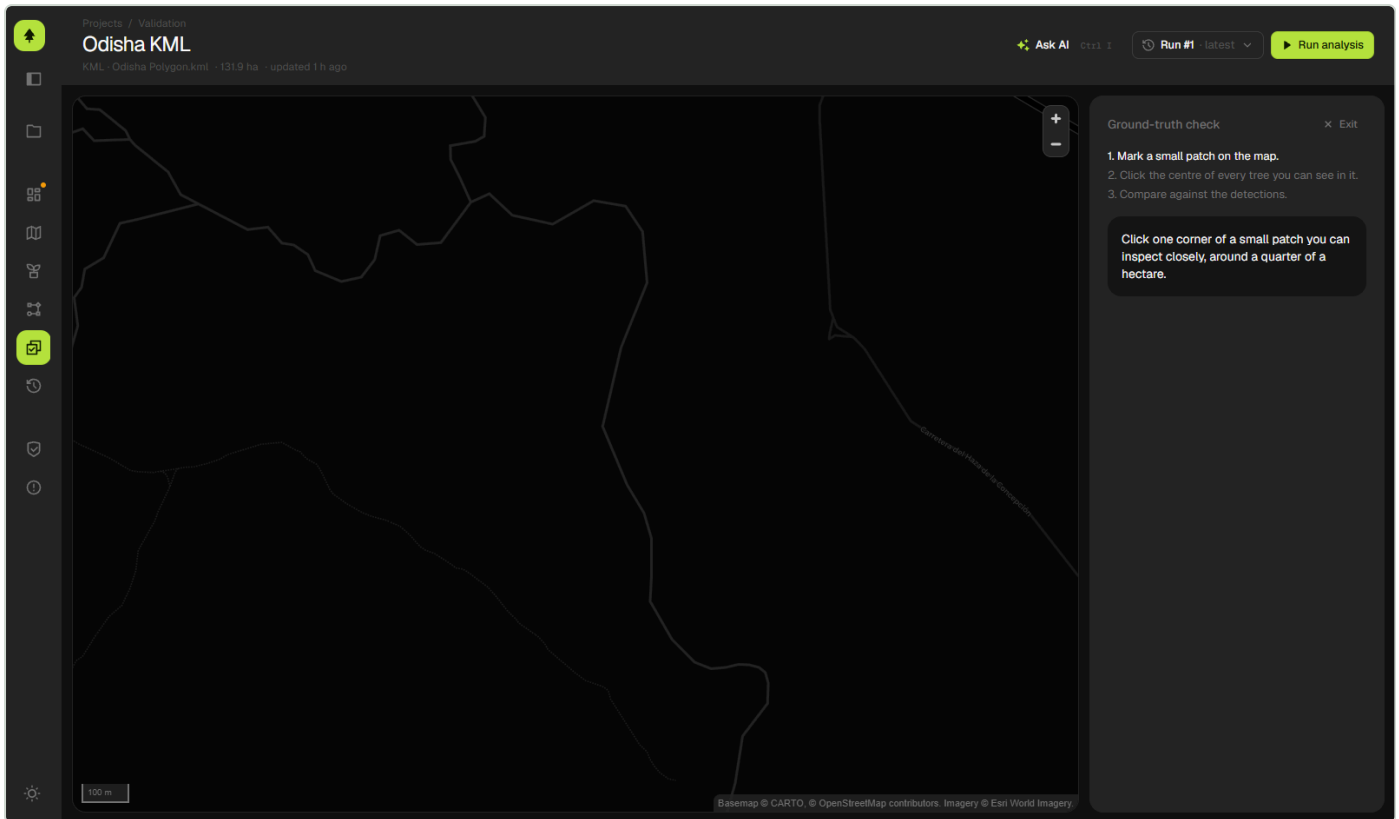
From this run

Numbers at this stage

Blob radius range	1.5–6.0 m
Blob threshold	0.02
Blobs found	339886

- The 7 stages: **source imagery** → **vegetation index** → **threshold** → **cleaned canopy mask** → **crown centres (or AI boxes)** → **crown segments** → **filtered and scored crowns**.
- Each stage is a real image from this run, with a **blend slider** to fade it over the imagery, its **method** in plain words, and its **numbers** (resolution, threshold, counts).
- Press **play** to walk through all stages automatically.

Say: "Most tools show the final map. This shows every intermediate step, so a reviewer can see exactly where a mistake would come from."



1. Mark a small patch on the map (about a quarter of a hectare).
2. Click the centre of every tree you can see in it; the detections are hidden so you are not biased.
3. CANOPY returns **precision, recall, F1**, matched / unmatched / missed trees, a **correction factor**, an adjusted count for the whole area, and optionally a small **local calibration model**.

Say: "On a new site you don't have to trust my numbers: two minutes of clicking gives you measured precision and recall there."

The screenshot shows a web interface for a project named 'Odisha KML'. The top bar includes a navigation menu on the left with icons for home, workspace, runs, and other functions. The main content area is titled 'Odisha KML' and shows a table of runs. The table has columns for Run, Status, Started, Changes, Cover, Crowns, and High conf. The first run, labeled '#1', is in 'Viewing' status, marked as 'Done', and started '1 h ago'. The 'Changes' column for this run shows 'Detector auto · Model balanced (s) · Min crown 3.0 m'. The 'Cover' is 56.3%, 'Crowns' are 17990-19904, and 'High conf.' is 52%. There are buttons for 'View' and 'Download' next to the first run. The interface also includes a 'Run history' section and a 'Run analysis' button.

Projects / Runs

Odisha KML

KML · Odisha Polygon.kml · 131.9 ha · updated 1 h ago

Ask AI Ctrl T Run #1 · latest Run analysis

Every run keeps its parameters and outputs

Run history

Run	Status	Started	Changes	Cover	Crowns	High conf.	
#1 Viewing	Done	1 h ago	Detector auto · Model balanced (s) · Min crown 3.0 m	56.3%	17990-19904	52%	View Download

- Every run of this workspace: status, when it started, **what changed** from the previous run (for example "Model balanced (s) → high accuracy (m)"), cover, crown range, and high-confidence share.
- View any old run, or download its bundle.

Say: "Change one setting, re-run, and the history shows exactly what moved the numbers."

Projects / Audit

Odisha KML

KML · Odisha Polygon.kml · 131.9 ha · updated 1 h ago

Ask AI Ctrl Z Run #1 · latest Run analysis

Exports

- Crowns** · GeoJSON
Crown polygons with every per-crown measurement and signal.
- Crowns** · CSV
The same crowns as a flat table for spreadsheets.
- Overlay** · PNG
Imagery with crown outlines coloured by confidence.
- Audit bundle** · ZIP
All outputs, manifest.json, summary and limitations.

Reproducibility

The same input and parameters produce a byte-identical `crowns.geojson`: JSON keys are sorted, coordinates rounded to 7 decimals (about 1 cm), and zip entries carry a fixed timestamp. Re-run any job and diff the files to check.

Job: `j_96e807c7`

Created: 2026-09-18T16:28:02Z

Manifest

Input	kml
Interpreted as	RGB, 3 bands, uint8 (Esri World Imagery JPEG tiles)
Imagery	Esri World Imagery, zoom 17
Fetched	2026-09-18T16:28:02Z
Resolution	1.11733 m/px
Resolution source	$156543.03392804097 * \cos(\text{centroid latitude}) / 2^{**}\text{zoom}$
Working CRS	EPSG:3857
Index p1-p99	0.0213 – 0.3651
Confidence weights	separation 0.31 · shadow 0.00 · shape 0.38 · size 0.31
AOI area	131.9 ha
Rejected regions	140
Edge crowns	309
Solar source	Height disabled

- **Exports:** crowns as GeoJSON and CSV, an overlay PNG, and the **audit bundle** zip (all outputs, manifest, summary, limitations).
- **Manifest:** input type, how it was interpreted, imagery source and zoom, fetch time, resolution and how it was computed, CRS, index range, confidence weights, area.
- **Reproducibility:** the same input and parameters give a byte-identical `crowns.geojson`.

Say: "An auditor can re-run this and get the same bytes. That's what makes a number usable in carbon work."

canopy

Workspace

- All projects 2

Recent

- Image Test
- Odisha KML

Evidence

- Limitations**

YOLO11 + classical CV
Every number ships with its provenance. No carbon claims.

What this tool cannot do
Plain-language limits of every number CANOPY reports.

Why this tool does not estimate carbon

Converting crown area to biomass requires species-specific allometric equations, wood density values, local calibration plots, and a root-to-shoot ratio. This tool has none of these, so it does not output tonnes of CO₂, biomass, or credit figures, not even roughly.

A carbon figure built on these outputs would inherit every error listed here, add the uncertainty of the conversion, and look far more precise than it is.

01 Merged crowns
In closed canopy, adjacent trees touching each other are often detected as one crown. This causes undercounting, and it is the main source of error in dense forest.

02 Understorey invisibility
Anything growing beneath the canopy is not visible from above and is not counted. In multi-layer forest this can be a large fraction of all stems.

03 Vegetation is not trees
With RGB imagery there is no reliable way to distinguish a tree from a tall hedge, a green roof, a crop field, or a shrub. The tool counts green blobs of roughly tree size.

04 No species, no age, no health
These require spectral bands, ground survey, or both.

05 Imagery date is often unknown
Basemap tiles carry no reliable acquisition date. The forest may have changed since capture.

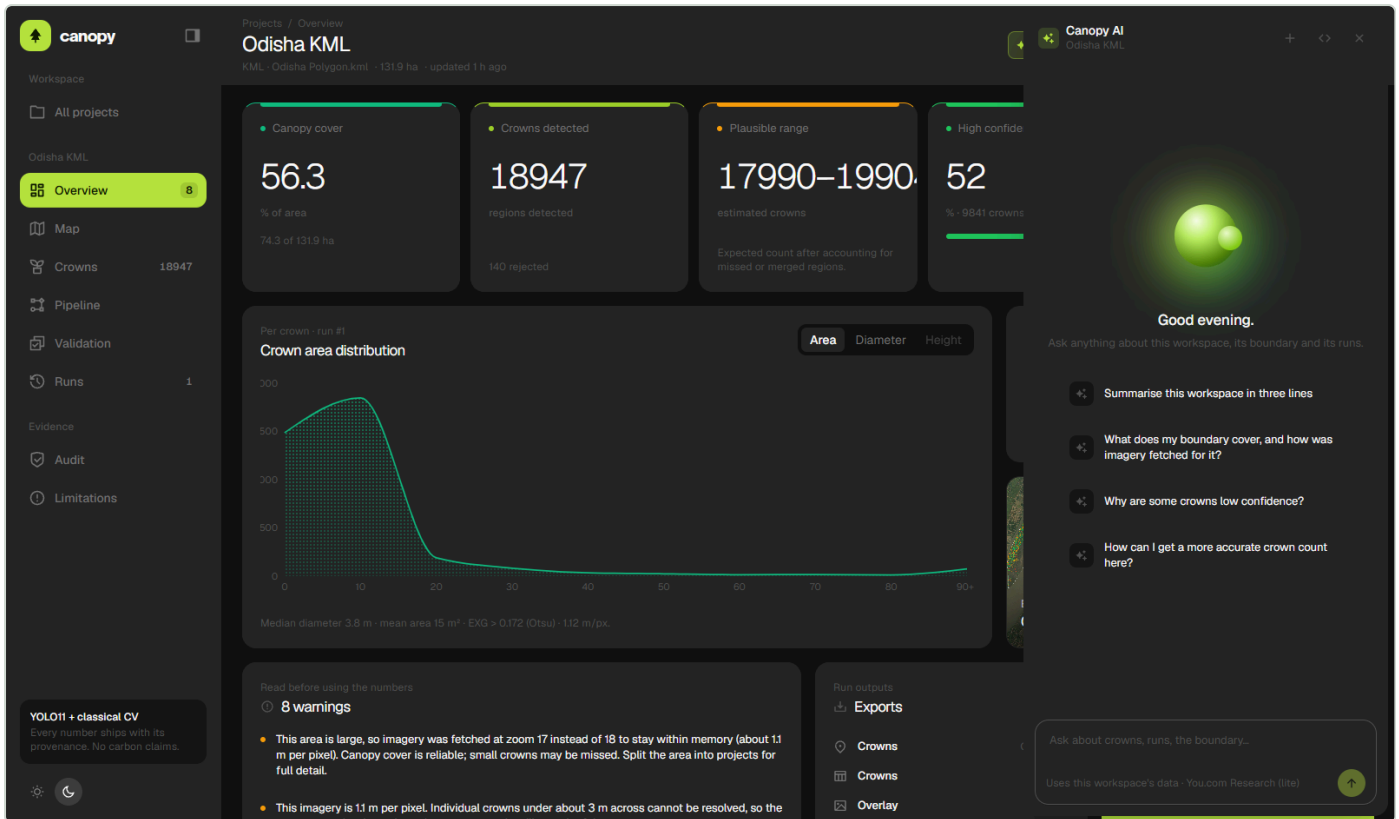
06 Shadow heights are geometric estimates
They assume flat terrain and an unobstructed shadow. On slopes and in dense canopy they are unreliable, which is why they are only reported for crowns passing the quality gate. The visible shadow is measured from the crown edge, so heights are biased low by roughly one crown radius times the tangent of the sun elevation.

07 Threshold sensitivity
Move the threshold slider and watch every number change. The tool reports the result of a choice, not a ground truth.

08 Local calibration needs labels
The optional local model only runs after a validation patch has enough matched and unmatched examples. Its estimate is specific to that imagery and patch; it must not be treated as a general model of trees or transferred to another scene.

A plain list of what CANOPY cannot do: no carbon or biomass figures, merged crowns in dense canopy, understorey hidden under the canopy, unknown imagery dates, and height only when the sun position is known.

Say: "I'd rather tell you what it can't do than show a number I can't back up."



- A panel slides in from the right. It knows this workspace: its boundary, its runs, its numbers and its warnings.
- Suggested questions: summarise the workspace, explain the boundary and imagery, why some crowns are low confidence, how to get a more accurate count.
- When it suggests a better setting, an **Apply and run** card appears; nothing runs without your click. Web facts come with numbered source links.
- Runs on the You.com Research API (lite tier, about 1 cent per question).

Say: "The assistant reads this workspace's own data first, so it explains these numbers rather than giving generic advice."